

SARAJEVO-BJELAVE, Bosnia and Herzegovina

WMO: 146540

Lat: 43.8678N

Lon: 18.4228E

Elev: 639

StdP: 93.88

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.6	-15.4	1.1	-10.0	-12.7	1.4	-7.2	6.5	6.8	5.8	4.6	1.4	110	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	33.2	19.8	31.3	19.4	29.4	18.9	21.7	29.7	20.5	28.5	19.5	27.3	2.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.8	26.5	17.7	13.7	24.3	16.8	12.9	22.8	66.3	29.9	61.8	28.4	58.3	27.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.7	4.9	4.2	-14.3	36.1	3.6	1.6	-16.9	37.2	-19.0	38.2	-21.0	39.1	-23.5	40.3		
(5)			-14.8	23.2	3.4	1.6	-17.2	24.3	-19.2	25.2	-21.0	26.1	-23.5	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	0.5	2.2	6.5	11.5	15.2	18.8	21.2	21.4	16.5	12.1	6.8	1.3		
	DBStd	8.55	5.23	5.36	4.94	3.93	3.79	4.01	3.50	3.57	4.01	4.20	5.02	4.86		
	HDD10.0	1104	297	221	128	29	6	1	0	1	3	29	118	272		
	HDD18.3	2909	554	452	368	207	110	42	14	13	81	196	345	527		
	CDD10.0	1544	1	3	18	73	166	266	347	354	197	92	23	3		
	CDD18.3	308	0	0	0	2	12	57	103	109	25	1	0	0		
	CDH23.3	2970	0	0	1	32	141	557	999	1003	222	17	0	0		
	CDH26.7	1124	0	0	0	3	26	181	416	429	71	0	0	0		
(14) Wind	WSAvg	1.7	1.5	1.7	2.0	1.9	1.7	1.7	1.7	1.7	1.6	1.6	1.7	1.5		
(15) Precipitation	PrecAvg	920	68	66	66	75	76	86	77	66	79	81	91	92		
	PrecMax	1252	159	183	174	140	203	161	190	158	258	259	179	294		
	PrecMin	628	4	13	1	8	28	23	8	13	6	1	18	1		
	PrecStd	152	39	40	36	27	37	37	39	35	47	52	42	53		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	16.3	21.8	26.8	29.4	32.9	35.4	36.1	32.3	25.3	20.1	14.9		
		MCWB	9.1	9.0	10.3	13.7	17.7	20.5	20.5	20.2	18.4	15.4	13.8	11.1		
	2%	DB	12.0	13.4	18.6	23.4	26.8	30.7	33.3	33.4	28.5	22.8	17.7	12.9		
		MCWB	7.8	7.7	9.4	12.9	16.2	19.4	19.9	19.7	17.8	14.8	12.0	8.9		
	5%	DB	9.8	11.7	16.2	20.7	24.6	28.8	31.2	31.2	25.5	20.6	16.1	11.2		
		MCWB	6.6	7.1	8.8	11.8	15.2	18.6	19.6	19.3	16.5	13.8	10.9	7.6		
	10%	DB	7.7	9.9	14.1	18.4	22.5	26.8	28.9	29.0	23.2	18.5	14.2	9.0		
		MCWB	5.0	6.0	8.1	10.7	14.5	17.7	19.1	18.8	15.7	13.0	9.8	6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.9	11.5	15.5	19.3	22.8	23.8	23.7	20.9	16.7	14.6	11.3		
		MCDB	12.9	15.1	18.1	23.8	27.2	29.9	32.8	33.0	28.2	22.9	19.3	14.6		
	2%	WB	8.1	8.3	10.3	13.8	17.4	20.6	21.5	21.6	18.4	15.3	12.5	9.3		
		MCDB	11.6	12.6	16.8	21.3	24.2	28.3	29.9	29.5	25.4	20.6	16.8	12.5		
	5%	WB	6.7	7.3	9.4	12.6	16.1	19.4	20.4	20.2	17.2	14.4	11.3	7.8		
		MCDB	9.6	11.2	15.1	19.0	22.3	26.9	28.7	28.1	24.0	19.3	15.6	10.9		
	10%	WB	5.2	6.2	8.4	11.5	15.0	18.3	19.4	19.2	16.2	13.5	10.1	6.3		
		MCDB	7.5	9.7	13.7	17.0	21.1	25.2	27.3	27.0	22.1	17.8	13.8	8.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.3	9.3	10.5	11.2	11.5	12.5	13.2	10.9	10.0	7.5	5.3		
		MCDBR	8.0	10.2	13.8	15.2	15.1	15.0	16.3	16.9	15.2	13.6	9.9	6.9		
	5% WB	MCWBR	5.1	6.0	6.9	7.0	6.4	5.9	6.2	6.3	6.5	6.7	5.7	4.9		
		MCWBR	7.5	8.9	11.5	13.1	13.1	13.7	14.6	14.6	13.1	11.9	8.6	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.341	0.385	0.438	0.434	0.426	0.427	0.428	0.404	0.370	0.333	0.311		
		taud	2.394	2.282	2.217	2.105	2.167	2.240	2.237	2.233	2.279	2.369	2.421	2.413		
	Ebn at Noon		810	840	845	825	840	847	841	826	816	798	776	773		
		Edh at Noon	77	102	124	151	146	136	135	131	116	93	75	69		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.13	3.31	4.23	5.23	6.00	6.21	5.60	3.90	2.71	1.66	1.22		
	RadStd		0.21	0.26	0.44	0.55	0.47	0.52	0.48	0.53	0.39	0.31	0.20	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	-106	+112	N/A

Nomenclature: See separate page